

1 Solution — GIAM 7.2

Problem 7

1.1 Problem

State necessary and sufficient conditions for the existence of an *Eulerian path* in a graph.

1.2 Definitions

A finite (multi-)graph $G = (V, E)$ has an *Eulerian path* if there is an *open walk* — one with distinct endpoints — that traverses every edge of G exactly once. The contrast with the Eulerian *circuit* of Problem 6 is the openness: here we demand the walk's start and end be different vertices.

1.3 Theorem (Eulerian Path)

A finite (multi-)graph G with at least one edge has an Eulerian path from u to v (with $u \neq v$) if and only if both of the following hold:

- (C1) G has *exactly two* vertices of odd degree, namely u and v .
- (C2) All edges of G lie in a single connected component.

When G is connected (so (C2) is automatic) and we ask merely whether *some* Eulerian path exists, the criterion collapses to:

A connected graph has an Eulerian path iff exactly two of its vertices have odd degree; the path must run between them.

Combined with Problem 6, the full picture for a connected graph with ≥ 1 edge is:

# odd-degree vertices	Eulerian feature
0	Eulerian circuit (closed walk)

2	Eulerian path (open, between the two odd)
4, 6, 8, ...	Neither

Table 1.1

(Counts other than 0, 2, 4, ... are impossible by Problem 2 — the number of odd-degree vertices is always even.)

1.4 Proof of Necessity ($\Rightarrow$)

Suppose G has an Eulerian path P from u to v with $u \neq v$.

Even degrees at interior vertices. Fix any vertex $w \notin \{u, v\}$. Each visit by P to w uses two distinct edges — one to enter and one to leave — and these incidences are paired up by the visits. Since P uses every edge of G exactly once, the edges at w are exactly the paired-up ones from the visits, so $\deg(w)$ is even.

Odd degrees at the endpoints. Now consider u (the start). The path leaves u once at the outset, then makes some number $k \geq 0$ of “visits” to u in the interior (each a pair: enter + leave). The total edges used at u is

$$1 + 2k,$$

which is *odd*. The same argument for v (using the *terminal* arrival as the unpaired incidence) gives $\deg(v)$ odd.

So the odd-degree vertices of G are *exactly* $\{u, v\}$ — establishing (C1).

Edge-connectivity (C2). A walk cannot jump between components: every two consecutive edges of P share a vertex, so all edges of P lie in a single connected component. Since P uses every edge, all edges of G share one component. $\square$

1.5 Proof of Sufficiency ($\Leftarrow$) — The Phantom-Edge Trick

Suppose (C1) and (C2) hold, with u, v the two odd-degree vertices of G . We reduce to the Eulerian-circuit theorem of Problem 6.

Step 1 — Augment G by a phantom edge. Build a new graph $G' = G + e_{uv}$ by adjoining a single new edge e_{uv} between u and v . (Even if G already has edges between u and v , simply add one more, parallel to the others.) In G' :

- $\deg_{G'}(u) = \deg_G(u) + 1$: was odd, now even.
- $\deg_{G'}(v) = \deg_G(v) + 1$: was odd, now even.
- All other vertices have unchanged degree, still even.
- Connectivity carries over: adding an edge to a connected graph keeps it connected.

So G' satisfies the hypotheses of Problem 6's theorem (all even degrees, edge-connected).

Step 2 — Get an Eulerian circuit of G' . By Euler's theorem (Problem 6), G' has an Eulerian circuit C . The circuit traverses every edge of G' exactly once — including the phantom edge e_{uv} .

Step 3 — Rotate so e_{uv} is last. A circuit can be re-parameterized to start at any of its edges; in particular, choose the parametrization so that e_{uv} is the *last* edge traversed. The circuit now looks like

$$v_0 \xrightarrow{e_1} v_1 \xrightarrow{e_2} v_2 \rightarrow \cdots \rightarrow v_{k-1} \xrightarrow{e_{uv}} v_0,$$

where the final edge $e_k = e_{uv}$ connects v_{k-1} and v_0 . By definition of e_{uv} , we must have $\{v_{k-1}, v_0\} = \{u, v\}$ — say WLOG $v_0 = v$ and $v_{k-1} = u$.

Step 4 — Remove the phantom edge. Delete the last edge e_{uv} from the circuit. What remains is the open walk

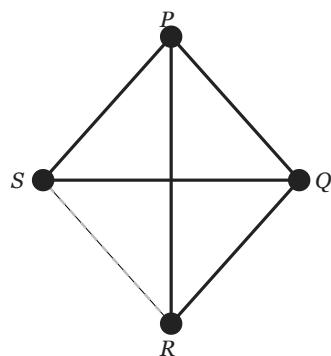
$$v \xrightarrow{e_1} v_1 \xrightarrow{e_2} \cdots \rightarrow v_{k-1} = u$$

— starting at v , ending at u , distinct endpoints — and using every edge of $G' \setminus \{e_{uv}\} = G$ exactly once.

This is an Eulerian path from v to u in G . ■

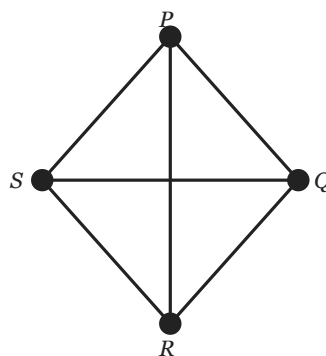
1.6 Examples

Has Eulerian path: K_4 minus edge RS



degrees: $P=3, Q=3, R=2, S=2$ — exactly 2 odd ✓

No Eulerian path: K_4



degrees: $3, 3, 3, 3$ — four odd vertices ✗

Figure 1.1: Left: K_4 with edge RS removed has degrees $(3, 3, 2, 2)$ — exactly two odd vertices P and Q — so an Eulerian path exists from P to Q . Right: K_4 has degrees $(3, 3, 3, 3)$ — four odd vertices — so no Eulerian path (and no circuit).

K_4 minus one edge (left). Removing the edge RS from K_4 yields a graph on $\{P, Q, R, S\}$ with five edges PQ, PR, PS, QR, QS and degree sequence $(3, 3, 2, 2)$. The two odd-degree vertices are P and Q , so by the theorem there is an Eulerian path from P to Q . One such path:

$$P \rightarrow R \rightarrow Q \rightarrow S \rightarrow P \rightarrow Q.$$

Edge check: PR, RQ, QS, SP, PQ — five edges, all distinct, starting at P and ending at Q . ✓

K_4 (right). All four vertices have degree 3 , so there are *four* odd-degree vertices. Since this is more than two, neither an Eulerian path nor an Eulerian circuit exists.

1.7 Königsberg Revisited

Euler's 1736 analysis of the Königsberg bridge problem actually proved this very theorem (and its circuit-version companion). The Königsberg graph has four vertices (river-banks plus two islands) of degrees $3, 3, 3, 5$ — *four* odd-degree vertices — so by Problem 7's theorem **no Eulerian path exists, even allowing distinct start and end points**. There is therefore no walking tour of Königsberg that crosses every bridge exactly once. This is the precise sense in which Euler “settled” the problem in the negative.

1.8 Remarks

- **Companion to Problem 6.** The two theorems share a single underlying parity argument:

$$\# \text{odd vertices} = \begin{cases} 0 & \iff \text{Eulerian circuit} \\ 2 & \iff \text{Eulerian path (open)} \\ \text{else} & \iff \text{neither} \end{cases}$$

Both proofs of necessity reduce to the visit-pairing observation: interior visits use two edges each; the two endpoints of an open path each contribute one *unpaired* incidence.

- **Phantom-edge construction is algorithmic.** Sufficiency is constructive: given (C1) and (C2), one can compute the Eulerian path in $O(|E|)$ time via the recipe above — add a phantom edge, run Hierholzer’s algorithm on the augmented graph, then delete the phantom edge from the circuit. (Implementation detail: rotate the circuit so the phantom edge sits at one end.)
- **Strict vs. loose convention.** Some authors use “Eulerian path” to mean *any* edge-once walk (circuit or open path). Under that convention, a graph has an Eulerian path iff it has 0 *or* 2 odd-degree vertices. The problem here (and the GIAM textbook) reserves “path” for the open case with distinct endpoints.
- **Generalization to digraphs.** For a directed graph, an Eulerian *directed* path from u to v exists iff the underlying graph is connected, $\deg^+(u) - \deg^-(u) = +1$, $\deg^+(v) - \deg^-(v) = -1$, and $\deg^+(w) = \deg^-(w)$ for every other vertex w .
- **Connection to Problem 1.** This is exactly the criterion applied to the four graphs in Problem 1: count the odd-degree vertices, then declare circuit / path / neither accordingly. The theorems of Problems 6 and 7 are the *justification* for that procedure.